ADDITIVITY OF THE CAP WITH A FIRST CHERN CLASS IN THE TENSOR PRODUCT OF TWO INVERTIBLE SHEAVES, AND VANISHING FOR THE TRIVIAL SHEAF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Additivity of the first Chern class of line bundles, on the level of the associated divisors. Divisors, label divisors-lemma-c1-additive (invoked at proof line 4345). Combined with the definitions of the cap with a first Chern class, it gives $c_1(\mathcal{L}) \cap \alpha + c_1(\mathcal{N}) \cap \alpha = c_1(\mathcal{L} \otimes \mathcal{N}) \cap \alpha$.
- The cap with the first Chern class of an invertible sheaf, defined through the divisor associated to a section. Same chapter chow.tex, the construction of the cap with c_1 for invertible sheaves (used at proof lines 4344-4346 and 4349). It is the definition the additivity statement is compared against, and it is what makes the vanishing for $\mathcal{O}_X$ follow from the behaviour of the constant section.

Source reference. Original lemma 02SL: sources/stacks-chow/tags.txt, source lines 3664–3664. Navigation pointer for \ref{divisors-lemma-c1-additive} used inside the counted span (line 4345); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 4331); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{L}$, $\mathcal{N}$ be an invertible sheaves on X. Then*

$$c_1(\mathcal{L}) \cap \alpha + c_1(\mathcal{N}) \cap \alpha = c_1(\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N}) \cap \alpha$$

in $\mathrm{CH}_k(X)$ *for every* $\alpha \in Z_{k+1}(X)$. *Moreover,* $c_1(\mathcal{O}_X) \cap \alpha = 0$ *for all* α.

Proof. The additivity follows directly from Divisors, Lemma 02SL and the definitions. To see that $c_1(\mathcal{O}_X) \cap \alpha = 0$ consider the section $1 \in \Gamma(X, \mathcal{O}_X)$. This restricts to an everywhere nonzero section on any integral closed subscheme $W \subset X$. Hence $c_1(\mathcal{O}_X) \cap [W] = 0$ as desired. □